

Lab Program 10

Title: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

File Structure:

prog10/

deployment.yaml

service.yaml

STEP 1: Start Minikube

\$ minikube start --cpus=2 --memory=4096

```
victus% minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
❗ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚢 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
```

STEP 2: Enable Kubernetes Dashboard

\$ minikube dashboard

```
victus% minikube dashboard
🐳 Verifying dashboard health ...
🚀 Launching proxy ...
🐳 Verifying proxy health ...
🌐 Opening http://127.0.0.1:42811/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser
```

STEP 3: Create deployment.yaml

\$ nano deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

STEP 3: Create a service.yaml

\$ nano service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

STEP 4: Apply resources

\$ kubectl apply -f deployment.yaml

```
victus% kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
```

\$ kubectl apply -f service.yaml

```
victus% kubectl apply -f service.yaml
service/nginx-service created
```

STEP 5: Monitor via CLI

\$ kubectl get pods

```
victus% kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-d4skg	1/1	Running	0	2m15s
nginx-deployment-6f9664446b-s87jc	1/1	Running	0	2m15s

STEP 6: Check Services

\$ kubectl get svc

```
victus% kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	7d10h
nginx-service	NodePort	10.104.91.40	<none>	80:31081/TCP	88s

STEP 7: Check pod logs

\$ kubectl logs nginx-deployment-6f9664446b-d4skg

```
victus% kubectl logs nginx-deployment-6f9664446b-d4skg
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:59:17 [notice] 1#1: using the "epoll" event method
2025/12/19 04:59:17 [notice] 1#1: nginx/1.29.4
```

STEP 8: Access the application via minikube and view in browser

\$ minikube service nginx-service

```
victus% minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31081



Opening service default/nginx-service in default browser...

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Kubernetes Dashboard

